

COURSE DESCRIPTION FORM: MT-1003 Calculus and Analytical Geometry

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAM TO BE EVALUATED BS-CS – Fall 2023

Course Description

Course Code	MT-1003		
Course Title	Calculus and Analytical Geometry		
Credit Hours	3		
Prerequisites by Course(s) and Topics	-		
Grading Policy	Absolute Grading		
Policy about missed and late assessment items in the course	Retake of missed assessment items (other than sessional / final exam) will not be held. A student who misses an assessment item (other than sessional / final exam) is awarded zero marks in that assessment item i.e., late submission will not be accepted. For missed sessional / final exam, exam retake/ pre-take application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee will decide the exam retake/ pre-take cases.		
Course Plagiarism Policy	Plagiarism in project or midterm/ final exam may result in F grade in the course. Plagiarism in an assessment item will result in zero marks in the respective assessment category.		
Attendance Policy	• Present – At the time of attendance. Attendance will be taken at the start of lecture. • Late – within first 10 minutes of the scheduled start of a lecture. Absent will be marked after 10 minutes. • Leaving the class for more than 5 minutes will result in absent status. Leaving the class without permission will result in absent status.		
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	100% Theory		
	Assessment items		
	Assessment Item	Number	Weight (%)
	Assignments	4-6	5
	Quiz	5-7	08
	Sessional I	1	15
	Sessional II	1	15
	Project	1	07
	Final Exam	1	50
Course Instructors	Dr. Hamda Khan, Mr. Usman Rashid		
Lab Instructors	-		
Course Coordinator	Dr. Hamda Khan		
URL (if any)	-		

Current Catalog Description	Topics including Function, Limits, Continuity, Curve Sketching, Derivatives, Chain Rule, Application of differentiation, Extreme values, Integration, Techniques of integration, Applications of integration, Transcendental Functions, L'Hôpital's rule, Inverse trigonometric functions, Sequence, Series (infinite series, geometric series), Divergence and convergence test for series will be covered in the course.			
Textbook	Thomas' Calculus, Early Transcendentals 13 th ed., by George B. Thomas Jr, Maurice D. Weir and Joel Hass, Pearson.			
Reference Material	1. Calculus with analytic geometry by Earl W. Swokowski, Alternate edition, Prindle, Weber & Schmidt. 2. Calculus Early Transcendentals, 8 th ed., by James Stewart, Cengage Learning.			
Course Learning Outcomes	A. Course Learning Outcomes (CLOs)			
	After completion of the course, the students shall be able to: 1. Apply knowledge of mathematics, science and appropriate knowledge domain to mathematically model and conceptualize defined problems. 2. Use fundamental principles of mathematics and relevant domains to identify, analyse and solve problems in order to reach substantiated conclusions. 3. Adapt modern computing tools such as MATLAB to apply appropriate mathematical techniques on complex computing activities. 4. Organize and prioritize work effectively as an individual or as a member/leader of a technical team.			
	B. Program Learning Outcomes			
	Covered attribute is indicated with a tick (✓) mark			
	PLO 1	1. Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.	✓
	PLO 2	2. Problem Analysis	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✓
	PLO 3	3. Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	
	PLO 4	4. Investigation & Experimentation	Conduct investigation of complex computing problems using research based knowledge and research based methods.	
	PLO 5	5. Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.	✓
	PLO 6	6. Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.	
PLO 7	7. Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems.		

	PLO 8	8. Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.	
	PLO 9	9. Individual and Teamwork:	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	✓
	PLO 10	10. Communication	Communicate effectively on complex computing activities with the computing community and with society at large.	
	PLO 11	11. Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team.	
	PLO 12	12. Life Long Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.	
	C. Mapping of CLOs on PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)			
			PLOs	
			1	2
			3	4
			5	6
			7	8
			9	10
			11	12
	CLOs	1	✓	
		2		✓
		3		
		4		
Topics covered in the course (assume 15-week instruction and 3 contact hours per week)	Topics			No. of Weeks
				Contact Hours
				CLOs
	- Properties of real numbers, functions, domain and range of functions, problem formulation, sketching of functions (trigonometric, logarithmic, inverse and exponential) - Use of modern tools / technology: sketching			1.5
	- Limits of function, calculating limits using limit laws, one sided limit, two-sided limit, continuity - Use of modern tools / technology: finding limits			4.5
	- Limits at Infinity; Horizontal and vertical Asymptotes and sketching of asymptotes - Use of modern tools / technology: check behavior of a function at infinity			1.5
	- Derivatives; slopes, tangents, derivative at a point, derivatives of function, differentiation rules - Use of modern tools / technology: using technology to solve problems involving derivatives			0.5
	Derivative as rate of change, derivative of trigonometric functions, chain rule, implicit differentiation, derivative of inverse logarithmic and inverse trigonometric functions			1
				3
				2
				6
				1,2,3,4

	• Use of modern tools / technology: using technology to solve problems involving derivatives			
	• More on application of differentiation; related rates, extreme values of functions, increasing/decreasing functions, derivative tests, local minimum/maximum, L'Hôpital's rule, concavity, curve sketching, procedure of applied optimization, modeling of problem • Use of modern tools / technology: using technology to solve problems involving derivatives • Linearization	3	9	1,2,3,4
	• Antiderivatives; problem formulation, initial value problem solution • Use of modern tools / technology: solution of initial value problem	0.5	1.5	1,2,3,4
	• Definite integrals, fundamental theorem of Calculus, substitution method for indefinite and definite integrals, area between curves • Use of modern tools / technology: using technology to calculate integrals	2	6	1,2,3,4
	• Integration by parts, trigonometric integrals, trigonometric substitution, improper integrals	1.5	4.5	1,2,3,4
	• Sequences, infinite series, geometric series, partial sum, nth-term test • Use of modern tools / technology: programming to check convergence or divergence	1.5	4.5	1,2,3,4
	Total	15	45	
Laboratory Projects/Experiments Done in the Course	-			
Programming Language for Assignments	MATLAB			
Class Time Spent (in hours – per week)	Theory (%)	Problem Analysis (%)	Solution Design (%)	Social and Ethical Issues (%)
	35	30	30	5
Oral and Written Communications	Every student is required to submit _01___ written report of typically _5 -10_ pages.			

COURSE CONTENTS			
Weeks	Contents/Topics	Courseware Events (Lab/ Case Study/ Quiz/ Assignment/ Project/ Presentation/ Research Report/ Term Paper etc.)	Comments (if any)
1	Introduction to calculus, importance of calculus in computing. Functions, domain-range, problem formulation. Even-odd functions, symmetry, common graphs, exercise 1.1. Combining functions, composition, translation.		
2	Scaling, sketching of functions (trig, log, exponential, inverse) exercise 1.2. Discussion on important functions for algorithm analysis. Definition of limit, left-hand right-hand limits, examples. Exercise 2.2.	Quiz # 1	
3	One-sided limits, exercise 2.4. Continuity, exercise 2.5. Limits involving infinity, asymptotes of graphs.	Assignment # 1	
4	Exercise 2.6. Use of limits in asymptotic analysis, slopes, tangents, derivative. Derivative at point, by definition, exercise 3.1. Use of derivatives, slopes & tangents in machine learning and signal processing. Derivatives of function, exercise 3.2, differentiation rules, exercise 3.3.	Quiz # 2	
5	Derivatives as rate of change, exercise 3.4. Derivatives of trigonometric functions, exercise 3.5.	Quiz # 3 Sessional I	
6	Chain rule; examples, exercise 3.6. Implicit differentiation; examples of real-life scenarios, exercise 3.7. Derivatives of inverse, logarithmic and inverse trigonometric functions, exercise 3.8, exercise 3.9.	Sessional I Assignment # 2	
7	Related rates, problem formulation, examples, exercise 3.10. Familiarization with advanced applications of differentiation in scenarios that use regression (weather forecasting and price prediction). Linearization, exercise 3.11.	Quiz # 4	
8	Extreme values of functions, examples, exercise 4.1. First derivative test, increasing decreasing functions, local minima, local maxima, examples and exercise 4.3, theorems. Concavity, examples.	Assignment # 3	

	Curve sketching.		
9	Exercise 4.4. L'Hôpital's rule, examples, exercise 4.5. Procedure of applied optimization, modeling of problem, examples.		
10	Exercise 4.6. Introduction to gradient descent and its applications. Antiderivatives, examples, exercise 4.8. The definite integral.	Assignment # 4 Quiz # 5	
11	Exercise 5.3. Fundamental theorem of calculus, exercise 5.4. Substitution method for indefinite integral, exercise 5.5.		
12	Substitution method for definite integral, area between curves, examples, exercise 5.6, exercise 8.1. Applications of integrals in computer science (computer games, terrain modeling, dealing with nondeterminism and uncertainty).	Sessional II Assignment # 5	
13	Integration by parts, trigonometric integrals, exercise 8.2, exercise 8.3. Trigonometric substitution, exercise 8.4.	Project Quiz # 6	
14	Integration of rational functions by partial fractions, exercise 8.5. Improper integrals, examples, exercise 8.8. Sequences, exercise 10.1.	Quiz # 7	
15	Infinite series, geometric series, partial sum, divergence test, exercise 10.2. Analyzing an algorithm with recursion. Revision.		